Decoupled Reprojection Consistency for Diagnosing 3D Gaussian Splatting Failures

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Abstract

*This paper introduces **Decoupled Reprojection Consistency (DRC)**, a training-free diagnostic for 3D Gaussian Splatting (3DGS) that decomposes an ambiguous reprojection error into three interpretable maps: a geometry/visibility mismatch E_{depth}, a base photometric inconsistency E_{base} computed under SH truncation (L=0), and a view-dependent residual E_{vd}. By comparing cross-view reprojection errors from zeroth-order (*base*) and trained-degree (*full*) SH renders of the same model, DRC separates geometry/visibility-driven inconsistency from appearance-driven instability. Paired with a bivariate fingerprint (quadrant occupancy) and coverage reporting, DRC turns reprojection consistency into actionable per-pixel triage. The method is demonstrated on synthetic and real scenes, separating glossy regions from geometry failures in our examples.*

CCS Concepts

• ***Computing methodologies*** → ***Rendering;*** *Image-based rendering;*

1. Introduction

Reprojection consistency—warping one view into another and measuring photometric agreement—is a standard sanity check for radiance-field reconstructions [WBF*17]. However, a single error map is ambiguous: elevated error can stem from geometry/visibility failures (depth errors, occlusions, floaters) *or* from view-dependent appearance that does not warp well (glossy highlights, reflections). In 3DGS [KKLD23], view dependence is encoded via per-Gaussian spherical harmonics (SH), so reflective regions may look plausible per view yet become inconsistent under cross-view warping. Existing training-free consistency tools [STGP26, GRS*24] typically produce a single scalar map per view pair—useful for ranking but ambiguous for debugging.

A simple internal ablation addresses this: rendering the trained model with SH truncated to L=0 (*base*) versus the trained SH degree (*full*), and comparing reprojection inconsistencies under both renders to obtain a cause-separating diagnostic axis. Note that L=0 is used solely as an internal basis truncation for diagnostics, not as a physical diffuse or albedo decomposition. Concretely, the contributions are: (1) a training-free tri-map decomposition into E_{depth} (geometry/visibility), E_{base} (SH0/base), and E_{vd} (view-dependent residual); (2) coverage-aware reporting with a bivariate fingerprint and quadrant labeling for actionable triage; and (3) demonstrations on synthetic and real scenes from public datasets.

2. Method

Inputs. A trained 3DGS model, calibrated cameras $\{K_i, R_i, t_i\}$, and per-view buffers $\{I_i^{\text{base}}, I_i^{\text{full}}, D_i, A_i\}$. The base render uses SH truncation to the DC term only (L=0); the full render uses the model's trained SH degree. In 3DGS [KKLD23], outgoing radiance for direction $\mathbf{d}$ is $\mathbf{c}(\mathbf{d}) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} \mathbf{c}_{\ell m} Y_\ell^m(\mathbf{d})$.

Neighbor selection and warping. For each target view i, neighbors $j \in \mathcal{N}(i)$ are selected via camera-center kNN. Pixels in i are back-projected using D_i to 3D, reprojected into each j, and bilinearly sampled from I_j^{base} and I_j^{full} to obtain warped images $\tilde{I}_j^{\text{base}\to i}$ and $\tilde{I}_j^{\text{full}\to i}$.

Masks and coverage. Two validity masks are used per pair (i, j): (a) M^{geo}: in-bounds $\wedge$ alpha gating—used for E_{depth}; and (b) M^{photo}: $M^{\text{geo}} \wedge$ depth-consistency—used for E_{base} and E_{full}, where photometric comparison is meaningful only when geometry agrees.

Photometric coverage C_i^{photo}, the fraction of pixels that have at least one valid photometric neighbor, is reported alongside every result.

Error maps. Let $\rho(\cdot,\cdot)$ be the L1 photometric error. For each target view i, inconsistencies across neighbors are robustly aggregated (median) on valid pixels:

- $E_{\text{depth}}^{(i)}(x)$: median relative depth mismatch $\left|D_j(\tilde{x}) - z^{\text{pred}}\right| / (z^{\text{pred}} + \epsilon)$ across j on M^{geo}.
- $E_{\text{base}}^{(i)}(x)$ and $E_{\text{full}}^{(i)}(x)$: median $\rho\left(I_i^{\text{base/full}}(x), \tilde{I}_j^{\text{base/full}\to i}(x)\right)$ across j on M^{photo}.
- $E_{\text{vd}}^{(i)}(x) = \max\left(0, E_{\text{full}}^{(i)}(x) - E_{\text{base}}^{(i)}(x)\right)$.

Normalization and quadrant labels. E_{base} and E_{full} are normalized by a shared per-view scale $s^{(i)} = q_{90}\left(\{E_{\text{base}}^{(i)}\} \cup \{E_{\text{full}}^{(i)}\}\right) + \epsilon$,

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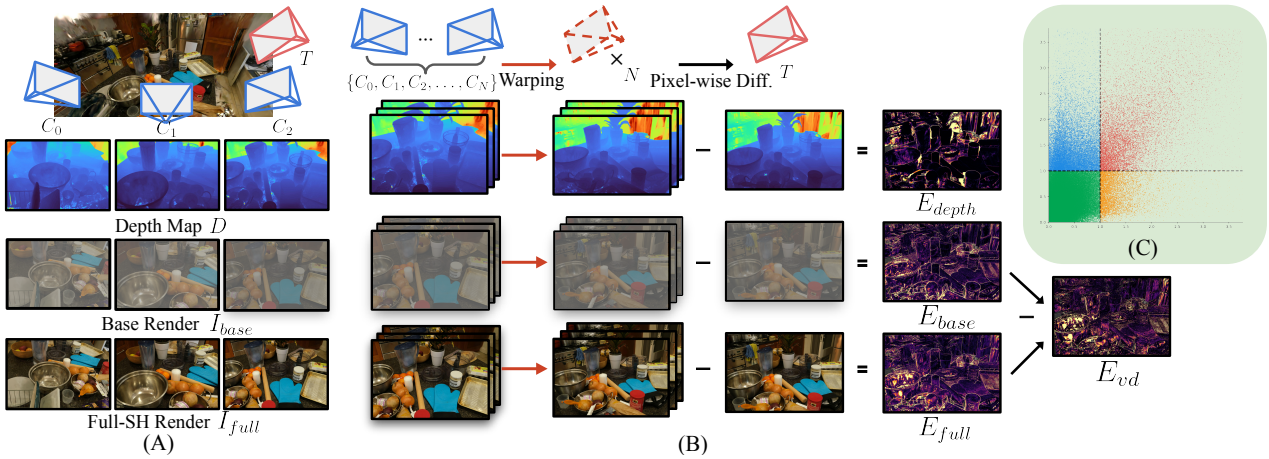


Figure 1: ***DRC pipeline.*** ***(A)*** *For target view T and context views $\{C_j\}$, render depth D, base color I_{base} (L=0), and full color I_{full} (trained SH degree).* ***(B)*** *Backward-warp context views into T using D_T; compute E_{depth}, E_{base}, E_{full}, and the residual $E_{\text{vd}}=\max(0, E_{\text{full}}-E_{\text{base}})$.* ***(C)*** *Per-view fingerprint: normalized $(E_{\text{base}}, E_{\text{full}})$ scatter with quadrant partitioning.*

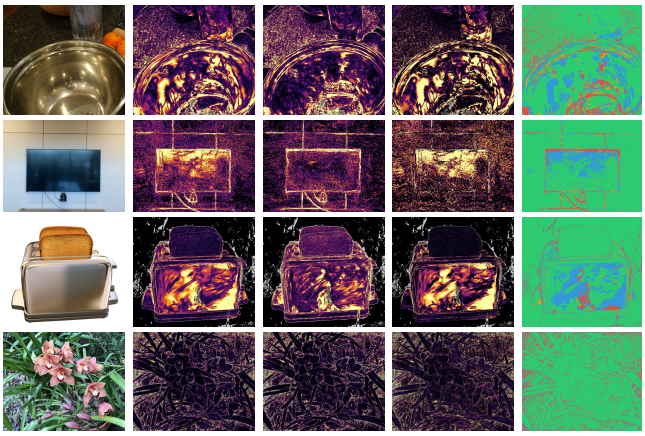


Figure 2: ***Qualitative triage.*** *Rows: Mip-NeRF 360* COUNTER *(stainless bowl ROI), LLFF* ROOM *(TV ROI), NeRF-Synthetic* TOASTER*, LLFF* ORCHIDS*. Columns: (a) RGB, (b) E_{full}, (c) E_{base}, (d) E_{vd}, (e) quadrant labels. Gray pixels are excluded by masking.*

computed on valid pixels, and define $E_{\bullet,n} = E_\bullet / s^{(i)}$. With threshold t=1, each valid pixel is assigned to one quadrant: *Consistent* (both low), *Geo/Vis-like* (both high), *Appearance-driven* ($E_{\text{base},n} \le t < E_{\text{full},n}$), or *Debug/Unexpected* ($E_{\text{base},n} > t \ge E_{\text{full},n}$).

3. Experiments

Experiments use Mip-NeRF 360 COUNTER, LLFF ROOM and ORCHIDS, and NeRF-Synthetic TOASTER. Fig. 2 shows that the single-map E_{full} conflates photometric mismatch and view-dependent effects: on COUNTER, marble elevates both E_{full} and E_{base}, while the stainless bowl is most prominent in E_{vd}; on ROOM and TOASTER, glossy surfaces (TV, toaster body) are labeled appearance-driven while edges fall into the Geo/Vis-like regime; on the matte ORCHIDS, appearance-driven pixels are sparse. Table 1 confirms high coverage ($\bar{C}_{\text{photo}} \approx 0.85$–$0.97$) and that the appearance-driven fraction is consistent with scene glossiness.

Scene (N)	$\bar{C}_{\text{photo}}$	Cons.	Geo	App	Dbg
M360/counter (216)	0.947	85.8	5.8	6.1	2.2
LLFF/room (37)	0.967	85.1	5.0	8.1	1.8
Syn/toaster (100)	0.854	85.0	5.0	8.2	1.8
LLFF/orchids (23)	0.852	87.7	7.7	3.1	1.4

Table 1: ***Coverage and quadrant occupancy on full target views.*** *$\bar{C}_{\text{photo}}$: mean photometric coverage. Cons./Geo/App/Dbg: quadrant occupancy (%) on valid pixels with t=1.*

4. Limitations and Takeaway

DRC is a diagnostic heuristic, not a ground-truth classifier: perfect mirrors and transparent glass can confound depth-based warping, and L=0 truncation is not a physical intrinsic decomposition. Low coverage from extreme pose gaps yields sparse maps; therefore, C^{photo} is always reported. Despite these caveats, DRC turns an ambiguous single error map into a training-free, coverage-aware tri-map diagnostic that separates geometry/visibility failures from view-dependent instability for practical 3DGS debugging.

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